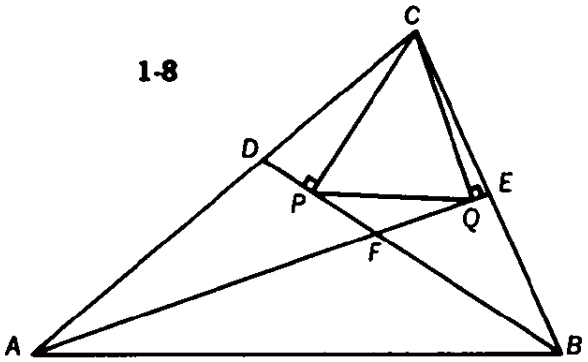
geometry-1-8

Two angle bisectors and two perpendiculars

Géométrie / Geometry | Level 3/5 (provisional)

Given any triangle ABC, $\overline{AE}$ bisects $\angle BAC$, $\overline{BD}$ bisects $\angle ABC$, $\overline{CP} \perp \overline{BD}$, and $\overline{CQ} \perp \overline{AE}$ (see figure), prove that $\overline{PQ}$ is parallel to $\overline{AB}$.



Challenging Problems In Geometry - 1-8; PDF p. 14; printed p. 3

Main numbered problem and original figure transcribed. Separately labelled Challenge not included.

Statement checked against source by Codex. Independent human review pending.